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Elektronika Analog BAB 8 NO. 1-30

1.

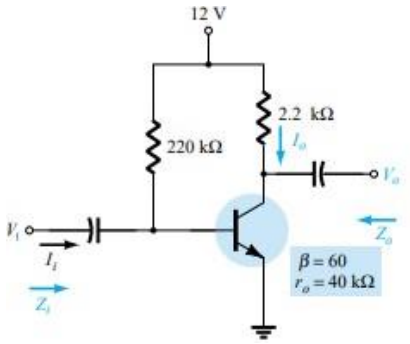


Figure 8.64

For the network of Fig. 8.64:

- Determine Z_i and Z_o.

re : IB = $\frac{Vcc-VBE}{RB} = \frac{12\,V - 07\,V}{220K\,\Omega} = 51.36$ µA

IE = (ß + 1) IB = (60 + 1)(51.36 µA) = 3.13 mA

re = $\frac{26\,mV}{IE} = \frac{26\,mV}{331\,mA}$ = 8.31 Ω

Zi = RB || ßre = 220K Ω || (60)(8.31 Ω) = 220K Ω || 498.6 Ω = **497.47 Ω**

ro ≥ 10Rc * Zo = Rc = **2.2K Ω**

- Find A_v and A_i

Av = - $\frac{Rc}{re} = \frac{-22K\,\Omega}{831\,\Omega}$ = **-264.74**

- Repeat part (a) with r_o = 20 kΩ.

Zi = **497.47 Ω**

Zo = ro || Rc = 20K Ω || 2.2K Ω = **1.98K Ω**

- Repeat part (b) with r_o = 20 kΩ.

Av = $\frac{-Rc \,||\, ro}{re} = \frac{-198K\,\Omega}{831\,\Omega}$ = **238.27**

2.

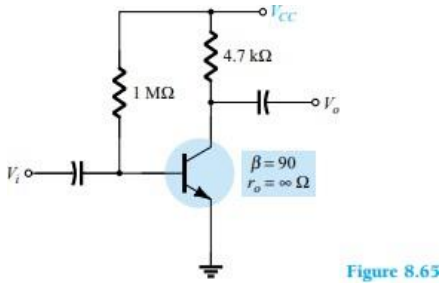


Figure 8.65

For the network of Fig. 8.65, determine V_{CC} for a voltage gain of A_v = - 200.

Av = - $\frac{Rc}{re}$ → re = - $\frac{Rc}{Av} = \frac{4.7K\,\Omega}{-200}$ = 23.5 Ω

re = $\frac{26\,mV}{IE}$ → IE = $\frac{26\,mV}{re} = \frac{26\,mV}{235\,\Omega}$ = 1.106 Ma

IB = $\frac{IE}{ß+1} = \frac{1106\,mA}{91}$ = 12.15 µA = IB = $\frac{Vcc-VBE}{RB}$

Vcc = IBRB + VBE = (12.15 µA)(1M Ω) + 0.7 V = 12.15 V + 0.7 V = **12.85 V**

3.

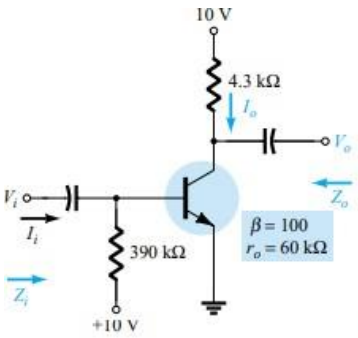


Figure 8.66

For the network of Fig. 8.66:

(a) Calculate I_B, I_C, r_e.

IB = $\frac{Vcc-VBE}{RB} = \frac{10\,V-07\,V}{390K\,\Omega} =$ **23.84 µA**

IE = (ß+1)IB=(101)(23.84 µA) = 2.40 mA

re = $\frac{26\,mV}{IE} = \frac{26\,mV}{189\,mA}$ = **13.76 Ω =** IC = ßIB = (100)(23.84 µA) = **2.384 mA**

(b) Determine Z_i and Z_o.

Zi = RB || ßre = 390K Ω || (100)(13.76 Ω) = **1.38K Ω**

Ro ≥ 10Rc *Zo = Rc = **5.6K Ω**

(c) Calculate A_v and A_i

Av = - $\frac{Rc}{re} = \frac{-56K\,\Omega}{1376\,\Omega}$ = **-406.98**

(d) Determine the effect of r_o = 30 kΩ on A_v and A_i.

Av = - $\frac{Rc\,\|r0}{re} = \frac{(56K\,\Omega)\|(30K\,\Omega)}{1376\,\Omega} = \frac{472K\,\Omega}{1376\,\Omega}$ = **343.03**

4.

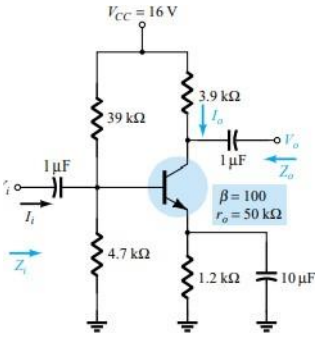


Figure 8.67 Problem 4

For the network of Fig. 8.67:

- Determine r_e.

VB = $\frac{R2\,VCC}{R1+R2} = \frac{47K\,\Omega\,(16\,V)}{39K\,\Omega+47K\,\Omega}$ = 1.721 V

VE = VB-VBE = 1.721 V – 0.7 V = 1.021 V

IE = $\frac{VE}{RE} = \frac{1021\,V}{1.2K\,\Omega}$ = 0.8507 mA

re = $\frac{26\,mV}{IE} = \frac{26\,mV}{08507\,mA}$ = **30.56 Ω**

- Calculate Z_i and Z_o.

Zi = R1 || R2 || ßre = 4.7K Ω ||39K Ω || (100)930.56 Ω) = **1.77K Ω**

ro ≥ 10Rc *Zo = Rc = **3.9K Ω**

- Find A_v and A_i.

Av = - $\frac{Rc}{re} = -\frac{39K\,\Omega}{3056\,\Omega}$ = **-127.6**

- Repeat parts (b) and (c) with r_o = 25 kΩ.

Zi = **1.77K**

Zo = Rc || ro = 3.9K Ω || 25K Ω = **3.37K Ω**

Av = - $\frac{(Rc \| r0)}{re}$ = - $\frac{(3.9K\ \Omega)\|(25K\ \Omega)}{3056\ \Omega}$ = - $\frac{337K\ \Omega}{3056\ \Omega}$ = **-110.28**

5.

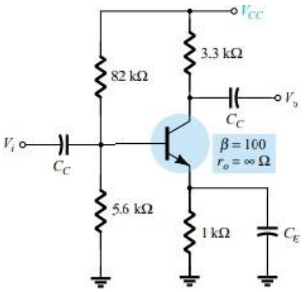


Figure 8.68 Problem 5

Determine V_{CC} for the network of Fig. 8.68 if A_v = −160 and r_o = 100 kΩ.

Av = = - $\frac{Rc}{re}$ = re = = - $\frac{Rc}{Av}$ = = - $\frac{33K\ \Omega}{-160}$ = **20.625 Ω**

re = $\frac{26\,mV}{IE}$ = IE = = - $\frac{26\,mV}{re}$ = $\frac{26\,mV}{20625\ \Omega}$ = 1.261 mA

IE = $\frac{VE}{RE}$ = VE =IERE = (1.261 mA)(1K Ω) = 1.261 V

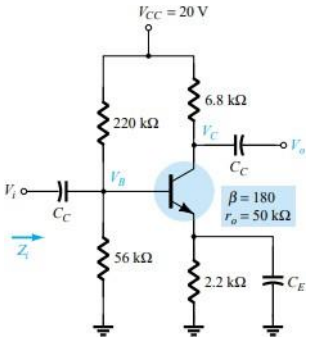
VB = VBE + VE = 0.7 V + 1.261 V = 1.961 V

VB = $\frac{56K\ \Omega\ VCC}{56K\ \Omega+82K\ \Omega}$ = 1.961 V

5.6K Ω VCC = (1.961 V)(87.6K Ω)

VCC = **30.68 V**

6.



For the network of Fig. 8.69:

- Determine Z_i.

  Rth = 56K Ω || 220K Ω = 44.64K Ω

  Eth = $\frac{56K\Omega(20V)}{220K\ \Omega+56K\ \Omega}$ = 4.058 V

  IB = $\frac{Eth-VBE}{Rth+(ß+1)RE}$ = $\frac{4058\ V-07\ V\ 4464K}{\Omega+(181)(22K\ \Omega)}$ = 7.58 µA

  IE = (ß + 1)IB = (181)(7.58 µA) = 1.372 mA

  re = $\frac{26\,mV}{IE}$ = $\frac{26\,mV}{1372\,mA}$ = **18.95 Ω**

- Calculate V_B and V_C.

  VE = IERE=(1.372 mA)(2.2K Ω) = 3.02 V

  VB = VE + VBE = 3.02 V + 0.7 V = 3.72 V

  VC = VCC – ICRC = 20V – ßIBRC = 20 V – (180)(7.58 µA)(4.7K Ω) = **13.59 V**

- Determine Z_i and A_v = $\frac{V_o}{V_i}$

  Zi = R1 || R2 || ßre

= 56K Ω || 220K Ω || (180)(7.58 μA) = 44.64K Ω || 3.14K Ω = **3.17K Ω**
ro < 10Rc

33.33K Ω < 10(4.7K Ω) = 47K Ω

Av = $-\frac{Rc \| r0}{re} = -\frac{(68K\ \Omega)\|(50K\ \Omega)}{18.95\ \Omega}$ = **-298.15**

7.

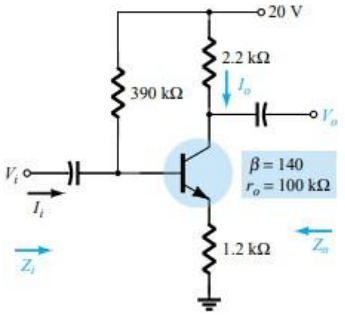


Figure 8.70 Problems 7 and 9

For the network of Fig. 870 :

- Determine I_B.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\ V - 0.7\ V}{390\ k\Omega + (141)(1.2\ k\Omega)} = \frac{19.3\ V}{559.2\ k\Omega} = 34.51\mu A$$

$$I_E = (\beta + 1)I_B = (140 + 1)(34.51\mu A) = 4.866mA$$

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{4.866mA} = 5.34\ \Omega$$

- Find Z_i and Z_o.

$Z_b = \beta r_e + (\beta + 1)R_E = (140)(5.34\ k\Omega) + (140 + 1)(1.2\ k\Omega) = 747.6 = 169.95\ k\Omega$

$Z_i = R_B \| Z_b = 390k\Omega \| 169.95\ k\Omega = 118.37\ k\Omega = R_C = \mathbf{2.2\ k\Omega}$

- Calculate A_v and A_i.

$A_v = -\frac{\beta R_c}{Z_b} = -\frac{(140)(22\ K\Omega)}{16995\ K\Omega} = \mathbf{-1.81}$

- Repeat parts (b) and (c) with $r_o = 20\ k\Omega$.

$$Z_b = \beta r_e + \left[\frac{(\beta+1) + R_C / r_o}{1 + (R_C + R_E)/r_o}\right]R_E$$

$$= 747.6\ \Omega \left[\frac{(141) + 2.2\ k\Omega / 20\ k\Omega}{1 + (3.4\ k\Omega)/20\ k\Omega}\right]1.2\ k\Omega$$

$$= 747.6\ \Omega + 144.72\ k\Omega$$

$$= 145.47\ k\Omega$$

$$Z_i = R_B \| Z_b = 390\ k\Omega \| 145.47\ k\Omega = \mathbf{105.95\ k\Omega}$$

$$Z_o = R_C = \mathbf{2.2\ k\Omega}\ \text{(any level of } r_o\text{)}$$

$$A_v = \frac{V_o}{V_i} = \frac{-\frac{\beta R_C}{Z_b}\left[1 + \frac{r_e}{r_o}\right] + \frac{R_C}{r_o}}{1 + \frac{R_C}{r_o}}$$

$$= \frac{\frac{-(140)(2.2\ k\Omega)}{145.47\ k\Omega}\left[1 + \frac{5.34\ \Omega}{20\ k\Omega}\right] + \frac{2.2\ k\Omega}{20\ k\Omega}}{1 + \frac{2.2\ k\Omega}{20\ k\Omega}}$$

$$= \frac{-2.117 + 0.11}{1.11} = \mathbf{-1.81}$$

8.

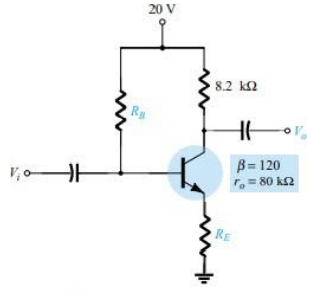


Figure 8.71 Problem 8

For the network of Fig. 8.71, determine R_E and R_B if $A_v = -10$ and $r_e = 3.8\Omega$. Assume that $Z_b = \beta R_E$.

$r_o \geq 10R_C$

$100\ \text{k}\Omega \geq 10(8.2\ \text{k}\Omega) = 82\ \text{k}\Omega$

Therefore:

$$A_v = -\frac{\beta R_C}{Z_b} = -\frac{\beta R_C}{\beta R_E} = -\frac{R_C}{R_E} = -10$$

$$\therefore R_E = \frac{R_C}{10} = \frac{8.2\ \text{k}\Omega}{10} = \mathbf{0.82\ k\Omega}$$

$$I_E = \frac{26\ \text{mV}}{r_e} = \frac{26\ \text{mV}}{3.8\ \Omega} = 6.842\ \text{mA}$$

$$V_E = I_E R_E = (6.842\ \text{mA})(0.82\ \text{k}\Omega) = 5.61\ \text{V}$$

$$V_B = V_E + V_{BE} = 5.61\ \text{V} + 0.7\ \text{V} = 6.31\ \text{V}$$

$$I_B = \frac{I_E}{(\beta+1)} = \frac{6.842\ \text{mA}}{121} = 56.55\ \mu\text{A}$$

$$\text{and } R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_B}{I_B} = \frac{20\ \text{V} - 6.31\ \text{V}}{56.55\ \mu\text{A}} = \mathbf{242.09\ k\Omega}$$

9. Repeat Problem 7 with R_E bypassed. Compare results.

dc analysis the same = r_e = 534 Ω

$Z_i = R_B \parallel Z_b = R_B \parallel \beta r_e = 390\ \text{k}\Omega \parallel (140)(5.34\ \Omega) = \mathbf{746.17}\ \Omega$ vs. 118.37 kΩ

$Z_o = R_C = \mathbf{2.2\ k\Omega}$ (as in #32)

$$= -\frac{R_C}{r_e} = -\frac{2.2\ \text{k}\Omega}{534\ \Omega} = \mathbf{-411.99}$$

$Z_i = \mathbf{746.17}\ \Omega$ vs. 105.95 kΩ for #19

$Z_o = R_C \parallel r_o = 2.2\ \text{k}\Omega \parallel 20\ \text{k}\Omega = \mathbf{1.98\ k\Omega}$ vs. 2.2 kΩ in #19

$$A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{1.98\ \text{k}\Omega}{5.34\ \Omega} = \mathbf{-370.79} \text{ vs. } -1.81 \text{ in #19}$$

10. 10.

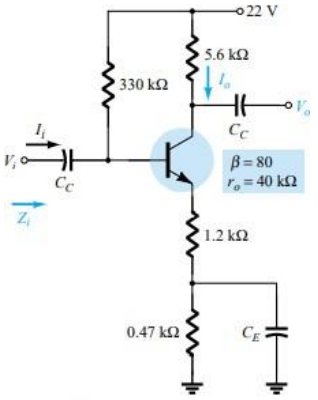


Figure 8.72 Problem 10

For the network of Fig. 872 :

Determine r_e.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{22\ V - 0.7\ V}{330\ k\Omega + (81)(1.2\ k\Omega + (0.47\ k\Omega))} = \frac{21.3\ V}{465.27\ k\Omega} = 45.78\mu A$$

$$I_E = (\beta + 1)I_B = (81)(45.78\mu A) = 3.71mA$$

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{3.71mA} = \mathbf{7}\ \Omega$$

Find Z_i and A_v.

$$r_o < 10(R_C + R_E)$$

$$\therefore Z_b = \beta r_e + \left[\frac{(\beta+1) + R_C/r_o}{1 + (R_C + R_E)/r_o}\right] R_E$$

$$= (80)(7\ \Omega) + \left[\frac{(81) + 5.6\ \text{k}\Omega/40\ \text{k}\Omega}{1 + 6.8\ \text{k}\Omega/40\ \text{k}\Omega}\right] 1.2\ \text{k}\Omega$$

$$= 560\ \Omega + \left[\frac{81 + 0.14}{1 + 0.17}\right] 1.2\ \text{k}\Omega$$

(note that $(\beta + 1) = 81 \gg R_C/r_o = 0.14$)

$$= 560\ \Omega + [81.14/1.17]1.2\ \text{k}\Omega = 560\ \Omega + 83.22\ \text{k}\Omega$$

$$= \mathbf{83.78\ k\Omega}$$

$$Z_i = R_B \| Z_b = 330\ \text{k}\Omega \| 83.78\ \text{k}\Omega = \mathbf{66.82\ k\Omega}$$

$$A_v = \frac{\dfrac{-\beta R_C}{Z_b}\left(1 + \dfrac{r_e}{r_o}\right) + \dfrac{R_C}{r_o}}{1 + \dfrac{R_C}{r_o}}$$

$$= \frac{\dfrac{-(80)(5.6\ \text{k}\Omega)}{83.78\ \text{k}\Omega}\left(1 + \dfrac{7\ \Omega}{40\ \text{k}\Omega}\right) + \dfrac{5.6\ \text{k}\Omega}{40\ \text{k}\Omega}}{1 + 5.6\ \text{k}\Omega/40\ \text{k}\Omega}$$

$$= \frac{-(5.35) + 0.14}{1 + 0.14}$$

$$= \mathbf{-4.57}$$

11. 11.

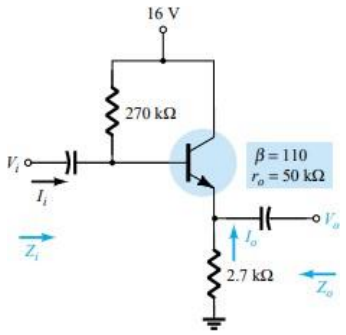


Figure 8.73 Problem 11

For the network of Fig. 873 :

- Determine r_e and βr_E.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{16\ V - 0.7\ V}{270\ k\Omega + (111)(2.7\ k\Omega)} = \frac{15.3\ V}{569.7\ k\Omega} = 26.86\mu A$$

$$I_E = (\beta + 1)I_B = (110 + 1)(26.86\mu A) = 2.98\ mA$$

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{2.98\ mA} = \mathbf{8.72\ \Omega} = \beta r_e = (110)(8.72\ \Omega) = \mathbf{959.2\ \Omega}$$

- Find Z_i and Z_o.

$Z_b = \beta r_e + (\beta + 1)R_E = 959.2\ \Omega + (111)(2.7\ k\Omega) = 747.6 = 300.66\ k\Omega$

$Z_i = R_B \| Z_b = 270\ k\Omega \| 300.66\ \text{k}\ \Omega = \mathbf{142.25\ k\Omega}$

$Z_o = R_E \| r_e = 2.7\ k\Omega \| 8.72\ \Omega = \mathbf{8.69\ \Omega}$

- Calculate A_v and A_i.

$$A_v = \frac{R_E}{R_E + r_e} = \frac{2.7\ \text{k}\Omega}{2.7\ \text{k}\Omega + 8.69\ \Omega} \cong \mathbf{0.997}$$

12. 12.

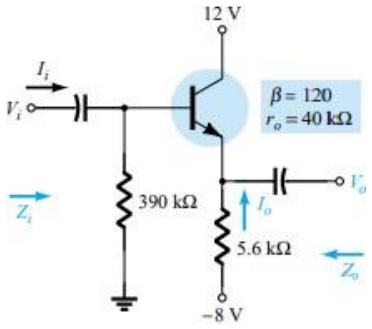


Figure 8.74 Problem 12

For the network of Fig. 874 :

- Determine Z_i and Z_o.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{8\,V - 0.7\,V}{390\,k\Omega + (121)(5.6\,k\Omega)} = 684\mu A$$

$$I_E = (\beta + 1)I_B = (121)(684\mu A) = 0.828\,mA$$

$$r_e = \frac{26\,mV}{I_E} = \frac{26\,mV}{0.828\,mA} = 31.4\,\Omega \rightarrow R_o = r_E = Z_b = 59816\,k\Omega$$

$Z_i = R_B || Z_b = 390\ k\Omega || 598.16\ \text{k}\ \Omega = 236.1\ \mathbf{k\Omega}$

$Z_o = R_E || r_e = 5.6\,k\Omega || 3.14\ \Omega = \mathbf{31.2}\ \Omega$

- Find A_v.

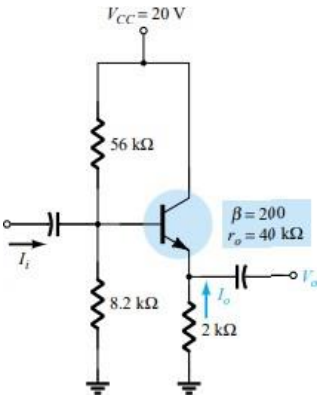
$$A_v = \frac{(\beta+1)R_E / Z_b}{1 + R_E / r_o} = \frac{(121)(5.6\ \text{k}\Omega)/598.16\ \text{k}\Omega}{1 + 5.6\ \text{k}\Omega / 40\ \text{k}\Omega} = \mathbf{0.994}$$

- Calculate V_o if V_i = 1 mV.

$$A_v = \frac{V_0}{V_i} = 0.994$$

$$V_o = A_v V_i = (0.994)(1\ \text{mV}) = \mathbf{0.994\ mV}$$

13. 13.



For the network of Fig. 875 :

- Calculate I_B and I_C.

Test $\beta R_E \geq 10R_2$

$(200)(2\ \text{k}\Omega) \geq 10(8.2\ \text{k}\Omega)$

$400\ \text{k}\Omega \geq 82\ \text{k}\Omega$ (checks)!

Use approximate approach:

$$V_B = \frac{8.2\ \text{k}\Omega(20\ \text{V})}{8.2\ \text{k}\Omega + 56\ \text{k}\Omega} = 2.5545\ \text{V}$$

$$V_E = V_B - V_{BE} = 2.5545\ \text{V} - 0.7\ \text{V} \cong 1.855\ \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.855\ \text{V}}{2\ \text{k}\Omega} = \mathbf{0.927\ mA}$$

$$I_B = \frac{I_E}{(\beta+1)} = \frac{0.927\ \text{mA}}{(200+1)} = \mathbf{4.61\ \mu A}$$

$$I_C = \beta I_B = (200)(4.61\ \mu\text{A}) = \mathbf{0.922\ mA}$$

- Determine r_e

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{0{,}927\ mA} = \mathbf{28{,}05\ \Omega}$$

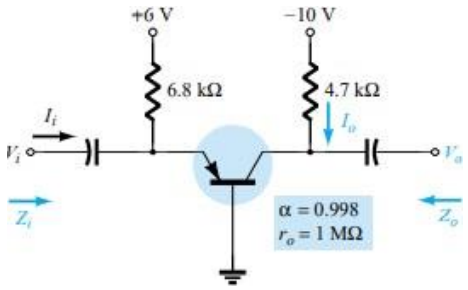
- Determine Z_i and Z_o

$$Z_b = \beta r_e + (\beta+1)R_E$$
$$= (200)(28.05\ \Omega) + (200+1)2\ \text{k}\Omega$$
$$= 5.61\ \text{k}\Omega + 402\ \text{k}\Omega = 407.61\ \text{k}\Omega$$
$$Z_i = 56\ \text{k}\Omega \parallel 8.2\ \text{k}\Omega \parallel 407.61\ \text{k}\Omega$$
$$= 7.15\ \text{k}\Omega \parallel 407.61\ \text{k}\Omega$$
$$= \mathbf{7.03\ k\Omega}$$
$$Z_o = R_E \parallel r_e = 2\ \text{k}\Omega \parallel 28.05\ \Omega = \mathbf{27.66\ \Omega}$$

- Find A_v and A_i.

$$A_v = \frac{R_E}{R_E + r_e} = \frac{2\ \text{k}\Omega}{2\ \text{k}\Omega + 28.05\ \Omega} = \mathbf{0.986}$$

14. 14.



For the common-base configuration of Fig. 8.76:

- Determine r_e

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6\ V - 0.7\ V}{(6.8\ k\Omega)} = 0.779\ mA$$

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{0{,}779\ mA} = \mathbf{33{,}38\ \Omega}$$

- Find Z_i and Z_o

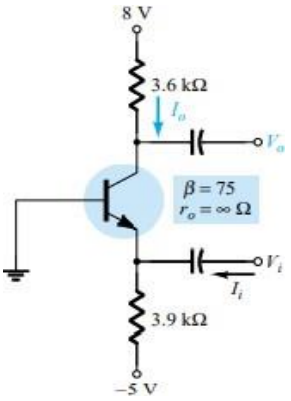
$$Z_i = R_E \parallel r_e = 6.8\ \text{k}\Omega \parallel 33.38\ \Omega$$
$$= \mathbf{33.22\ \Omega}$$

$$Z_o = R_C = \mathbf{4.7\ k\Omega}$$

- Calculate A_v and A_i.

$$A_v = \frac{\alpha R_C}{r_e} = \frac{(0.998)(4.7\ \text{k}\Omega)}{33.38\ \Omega}$$
$$= \mathbf{140.52}$$

15. 15.



For the network of Fig. 8.77, determine A_v and A_i.

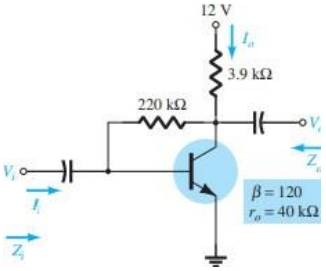
$$\alpha = \frac{\beta}{\beta+1} = \frac{75}{76} = 0.9868$$

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{5\ \text{V} - 0.7\ \text{V}}{3.9\ \text{k}\Omega} = \frac{4.3\ \text{V}}{3.9\ \text{k}\Omega} = 1.1\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{1.1\ \text{mA}} = 23.58\ \Omega$$

$$A_v = \alpha \frac{R_C}{r_e} = \frac{(0.9868)(3.9\ \text{k}\Omega)}{23.58\ \Omega} = \mathbf{163.2}$$

16. 16.



For the collector FB configuration of Fig. 8.78:

- Determine r_e

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{12\ V - 07}{220k\Omega + 120(3.9k\Omega)} = 16.42\mu A$$

$$I_E = (\beta + 1)I_B = (120 + 1)(16.42\mu A) = 1.987mA$$

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mmV}{1.987mA} = \mathbf{13.08\ \Omega}$$

- Find Z_i and Z_o

$$Z_i = \beta r_e \| \frac{R_F}{A_v}$$

$$A_v = \frac{-R_C}{r_e} = \frac{-3.9\ k\Omega}{13.08\Omega} = -298$$

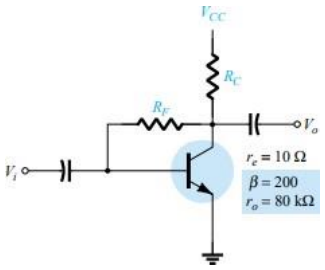
$$Z_i = (120)(13.08\Omega) \| \frac{220k\Omega}{298} = 1.5696\ k\Omega \| 738\Omega = \mathbf{501.98\ \Omega}$$

$$Z_o = R_C \| R_F = 3.9\ k\Omega \| 220k\Omega = \mathbf{3.83\ k\Omega}$$

- Calculate A_v and A_i

From above, $A_v = \mathbf{-298}$

17. 17.



Given $r_e = 10\Omega$, $\beta = 200$, $A_v = -160$, and $A_i = 19$ for the network of Fig. 8.79, determine R_C, R_F, and V_{CC}.

$$A_v = \frac{-R_C}{r_e} = -160$$

$$R_C = 160(r_e) = 160(10\Omega) = \mathbf{1.6\ k\Omega}$$

$$A_i = \frac{\beta R_F}{R_F + \beta R_C} = 19 \rightarrow 19 = \frac{200R_F}{R_F + 200(16\ \text{k}\Omega)} = R_F = \frac{3800R_C}{181} = \frac{3800(16\ \text{k}\Omega)}{181} = \mathbf{33.59\ k\Omega}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = I_B(R_F + \beta R_C) = V_{CC} - V_{BE} \quad \text{and} \quad V_{CC} = V_{BE} + I_B(R_F + \beta R_C)$$

with $I_E = \frac{26\ mV}{r_e} = \frac{26\ mV}{10\Omega} = 2.6\ mA \rightarrow I_B = \frac{I_E}{\beta + 1} = \frac{26mA}{200+1} = 12.94\ \mu A$

$$\therefore V_{CC} = V_{BE} + I_B(R_F + \beta R_C) = 0.7\ V + (12.94\ \mu A)(33.59\ k\Omega + (200)(16\ k\Omega))$$
$$= \mathbf{5.28\ V}$$

18. For the network of Fig. 8.30:

- Derive the approximate equation for A_v.

$A_v = V_o / V_i = I_b\beta r_e + (\beta+1) I_b R_E$

$I_o + I' = I_C = \beta I_b$, but $I_i = I' + I_b$ and $I' = I_i - I_b$

Substituting, $I_o + (I_i - I_b) = \beta I_b$ and $I_o = (\beta + 1)I_b - I_i$

Assuming $(\beta + 1)I_b \gg I_i \rightarrow I_o \cong (\beta + 1)I_b$ and $V_o = -I_o R_C = -(\beta + 1)I_b R_C$

Therefore, $A_v = \frac{V_o}{V_i} = \frac{-(\beta+1)I_b R_C}{I_b\beta R_E + (\beta+1)I_b R_E}$ and $A_v = \frac{V_o}{V_i} \cong -\frac{R_C}{r_e + R_E} = -\frac{R_C}{R_E}$

- Derive the approximate equation for A_i.
- Derive the approximate equations for Z_i and Z_o.

$V_i \cong \beta I_b (r_e + R_E)$ for $r_e \ll R_E \rightarrow V_i \cong \beta I_b R_E \rightarrow$ Now $I_i = I_b + I' + \frac{V_i - V_o}{R_F} + I_b$

Since $V_o \gg V_i \rightarrow I_i = -\frac{V_o}{R_F} + I_b$ or $I_i = I_b + \frac{V_o}{R_F}$ and $V_o = \beta R_E I_b + \beta \frac{V_o}{R_F} R_E$ but $V_o = A_v V_i$

and $V_i = \beta R_E I_b + \frac{\beta A_v V_i R_E}{R_F}$ or $V_i - \frac{A_v \beta R_E V_i}{R_F} = \beta R_E I_b$ $V_i\left[1 - \frac{A_v R_E}{R_F}\right] = [\beta R_E] I_i$

so $Z_i = \frac{V_i}{I_i} = \frac{\beta R_E}{1 - \frac{A_v \beta R_E}{R_F}} = \frac{\beta R_E R_F}{R_F + \beta(-A_v)R_E}$

$Z_i = \frac{V_i}{I_i} = x \| y$ where $x = \beta R_E$ and $y = R_F / |A_v|$

with $Z_i = \frac{x \cdot y}{x + y} = \frac{(\beta R_E)(R_F / |A_v|)}{\beta R_E + R_F / |A_v|}$

$$Z_i \cong \frac{\beta R_E R_F}{\beta R_E |A_v| + R_F}$$

Z_o: Set $V_i = 0$

$V_i = I_b\beta r_e + (\beta + 1)I_bR_E$

$V_i \cong \beta I_b(r_e + R_E) = 0$

since $\beta, r_e + R_E \neq 0 \quad I_b = 0$ and $\beta I_b = 0$

$$\therefore I_o = \frac{V_o}{R_C} + \frac{V_o}{R_F} = V_o\left[\frac{1}{R_C} + \frac{1}{R_F}\right]$$

$$\text{and } Z_o = \frac{V_o}{I_o} = \frac{1}{\frac{1}{R_C} + \frac{1}{R_F}} = \frac{R_CR_F}{R_C + R_F} = \boldsymbol{R_C \| R_F}$$

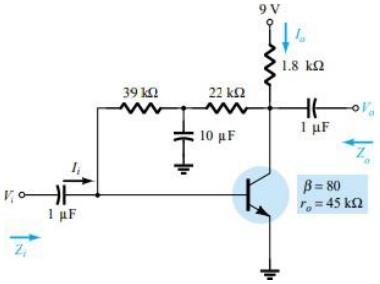
(d) Given $R_C = 2.2\text{k}\Omega$, $R_F = 120\ \text{k}\Omega$, $R_E = 1.2\ \text{k}\Omega$, $\beta = 90$, and $V_{CC} = 10\ \text{V}$, calculate the magnitudes of A_v, A_i, Z_i and Z_o using the equations of parts (a) through (c).

$$A_v \cong -\frac{R_C}{R_E} = -\frac{2.2\text{k}\Omega}{1.2\text{k}\Omega} = -1.83$$

$$Z_i \cong \frac{\beta R_E R_F}{\beta R_E|A_V| + R_F} = \frac{(90)(1.2\text{k}\Omega)(120\text{k}\Omega)}{(90)(1.2\text{k}\Omega)(1.83) + 120\text{k}\Omega} = \mathbf{40.8\ k\Omega}$$

$Z_o \cong R_C \| R_F = 2.2\text{k}\Omega \| 120\text{k}\Omega = \mathbf{2.16\ k\Omega}$

19. 19.



For the network of Fig. 8.80:

- Determine Z_i and Z_o.

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{9\ V - 0.7\ V}{(39\ k\Omega + 22k\Omega)(80)(1.8\ k\Omega)} = \frac{8.3\ v}{61\ k\Omega + 144\ k\Omega} = \frac{8.3\ V}{205\ k\Omega}$$

$$= 40.49\ \mu A$$

$$I_E = (\beta + 1)I_B = (80 + 1)(40.49\ \mu A) = 3.28\ mA$$

$$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{3.28\ mA} = 7.93\ \Omega$$

$Z_i = R_{F1} \| \beta r_e = 39\ \text{k}\Omega \| (80)(7.93\ \Omega) = 39\ \text{k}\Omega \| 634.4\ \Omega = 0.62\ \text{k}\Omega$

$Z_o = R_C \| R_{F2} = 1.8\ k\Omega \| 22\ \text{k}\Omega = \mathbf{1.66\ k\Omega}$

- Find A_v and A_i.

$$A_v = \frac{-R'}{r_e} = \frac{-R_C \| R_{F_2}}{r_e} = -\frac{1.8\ \text{k}\Omega \| 22\ \text{k}\Omega}{7.93\ \Omega}$$

$$= \frac{-1.664\ \text{k}\Omega}{7.93\ \Omega} = \mathbf{-209.82}$$

20. (a) Given $\beta = 120$, $r_e = 4.5\Omega$ and $r_o = 40\text{k}\Omega$, sketch the approximate hybrid equivalent circuit.

$h_{fe} = \beta = 120$

$h_{ie} \cong \beta r_e = (120)(4.5\Omega) = 540\ \Omega$

$$h_{oe} = \frac{1}{r_o} = \frac{1}{40\ k\Omega} = \mathbf{25\ \mu S}$$

(b) Given $h_{ie} = 1\ \text{k}\Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 90$, and $h_{oe} = 20\mu S$, skecth the r_e model.

$$r_e \cong \frac{h_{ie}}{\beta} = \frac{1\text{k}\Omega}{90} = 11.11\ \Omega$$

$\beta = h_{fe} = 90$

$$r_o = \frac{1}{h_{oe}} = \frac{1}{20\ \mu S} = \mathbf{50\ k\Omega}$$

21. For the network of Problem 1:

- Determine $r_e = 8.31\ \Omega$

(b) $h_{fe} = \beta = \mathbf{60}$

$h_{ie} = \beta r_e = (60)(8.31\ \Omega) = \mathbf{498.6\ \Omega}$

- Find Z_i and Z_o using the hybrid parameters.

$Z_i = R_B \| h_{ie} = 220\ \text{k}\Omega \| 498.6\ \Omega = \mathbf{497.47\ \Omega}$

$Z_o = R_C = \mathbf{2.2\ k\Omega}$

- Calculate A_v and A_i using the hybrid parameters.

$$A_v = \frac{-h_{fe}R_C}{h_{ie}} = \frac{-(60)(2.2k\Omega)}{4986\ \Omega} = \mathbf{-264.74}$$

$A_i \cong h_{fe} = \mathbf{60}$

- Determine Z_i and Z_o If $h_{oe} = 50\ \mu\text{S}$.

$Z_i = \mathbf{497.47\ \Omega}$ (*the same*)

$$Z_o = r_o \| R_C, \quad r_o = \frac{1}{25\ \mu S} = 40\ k\Omega \rightarrow 40\ k\Omega \| 2.2\ k\Omega = \mathbf{2.09\ k\Omega}$$

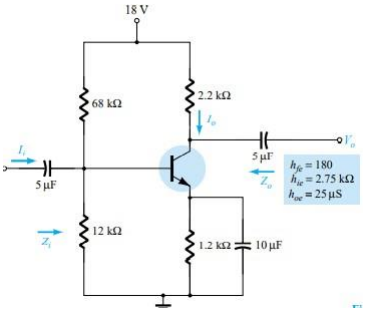
- Determine A_v and A_i If $h_{oe} = 50\ \mu\text{S}$.

$$A_v = \frac{-h_{fe}(r_o \| R_C)}{h_{ie}} = \frac{-(60)(2085\ k\Omega)}{4986\ \Omega} = \mathbf{-250.90}$$

$$A_i = -\frac{A_v Z_i}{R_C} = -\frac{(-250.90)(497.47\Omega)}{2.2\ k\Omega} = \mathbf{56.73}$$

- Compare the solutions above with those of Problem 1. (Note: The solutions are available in Appendix E if Problem 1 was not performed.)

22. 22.



For the network of Fig. 8.81 :

- Determine Z_i and Z_o.

68 kΩ || 12 kΩ = 10.2 kΩ

$Z_i = 10.2\ \text{k}\Omega \| h_{ie} = 10.2\ \text{k}\Omega \| 2.75\ \text{k}\Omega = \mathbf{2.166\ k\Omega}$

$Z_O = \text{RC} \| r_o = 2.2\ \text{k}\Omega \| 40\ \text{k}\Omega = \mathbf{2.085\ k\Omega}$

- Calculate A_v and A_i.

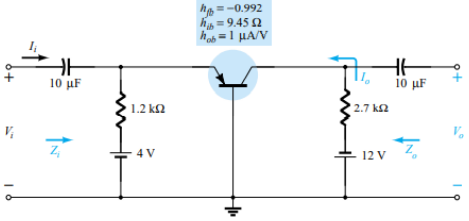
$$A_v = \frac{-h_{fe}R'_C}{h_{ie}} \rightarrow R'_C = R_C \| r_o = 2.085\ k\Omega$$

$$= \frac{-(180)(2.085\ k\Omega)}{2.75\ k\Omega} = \mathbf{-136.5}$$

$$A_i = \frac{I_O}{I_i} = \frac{I_O}{I'_i} \times \frac{I'_i}{I_i} = \mathbf{135.13}$$

- Determine r_e and βr_e to h_{ie}.

23. 23.



For the common-base network of Fig. 8.82:

(a) Deterime $Z_i = R_E \parallel h_{ib} = 1.2\ \text{k}\Omega \parallel 9.45\ \Omega = \mathbf{9.38\ \Omega}$

$Z_o = R_C \parallel \frac{1}{h_{ib}} = 2.7\ \text{k}\Omega \parallel \frac{1}{1\times10^{-6}A/V} = 2.7\ \text{k}\Omega \parallel 1\ \text{M}\Omega \cong \mathbf{2.7\ k\Omega}$

(b) $A_v = \dfrac{-h_{fb}(R_C \parallel \frac{1}{h_{ab}})}{h_{ib}} = \dfrac{-(-0992)(\cong 27\ \text{k}\Omega)}{945\ \ \Omega} = \mathbf{284.43}$

$A_i \cong \mathbf{-1}$

(c) Determine α, β, $r_e$and r_o.

$\alpha = -h_{fb} = -(-0.992) = \mathbf{0.992}$

$\beta = \frac{\alpha}{1-\alpha} = \frac{0992}{1-0992} = \mathbf{124}$

$r_e = h_{ib} = 9.45\ \Omega$

$r_o = \frac{1}{h_{ab}} = \frac{1}{1\mu A/V} = \mathbf{1\ M\Omega}$

24. Repeat parts (a) and (b) of Problem 22 with $h_{re} = 2\times10^{-4}$ and compare results. (a)

$$Z_i' = h_{ie} - \frac{h_{fe}h_{re}R_L}{1+h_{oe}R_L} = 2.75\ \text{k}\Omega - \frac{(180)(2\times10^{-4})(2.2\ \text{k}\Omega)}{(1+25\mu S)(2.2\ \text{k}\Omega)}$$
$$= 2.68\ \text{k}\Omega$$
$$Z_i = 10.2\ \text{k}\Omega \parallel Z_i' = \mathbf{2.12\ k\Omega}$$
$$Z_o' = \frac{1}{h_{oe} - (h_{fe}h_{re}/h_{ie})} = \frac{1}{25\ \mu S - (180)(2\times10^{-4})/2.75\ \text{k}\Omega}$$
$$= 83.75\ \text{k}\Omega$$
$$Z_o = 2.2\ \text{k}\Omega \parallel 83.75\ \text{k}\Omega = 2.14\ \text{k}\Omega$$

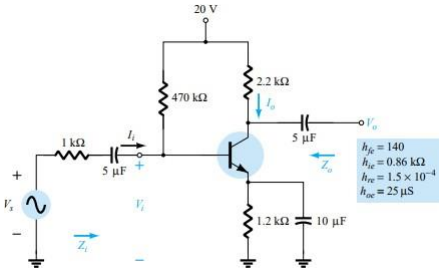
(b)

$$A_v = \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe}) - h_{fe}h_{re})R_L} = \frac{-(180)(2.2\ \text{k}\Omega)}{2.75\ \text{k}\Omega + \left((2.75\ \text{k}\Omega)(25\mu S) - (180)*2\times10^{-4}\right)2.2\ \text{k}\Omega}$$
$$= \mathbf{-140.3}$$

(c)

$$A_i' = \frac{h_{fe}}{1+h_{oe}R_L} = \frac{(180)}{1+(25\mu S)(2.2\ \text{k}\Omega)} = 170.62$$
$$A_i = \frac{I_o}{I_i} = \frac{I_o}{I_i'}\cdot\frac{I_i'}{I_i} = (170.62)\left(\frac{10.2\ \text{k}\Omega}{10.2\ \text{k}\Omega + 2.68\ \text{k}\Omega}\right)$$
$$= \mathbf{135.13}$$

25.



For the network of Fig. 8.83, determine:

(a) $Z_i' = h_i = \frac{h_{fe}h_{re}R_L}{1+h_{oe}R_L} = 0.86k\Omega - \frac{(140)(15x10^{-4})(22k\Omega)}{1+(25\ \mu S)(22k\Omega)} = 0.86k\Omega - 43.79\Omega = 816.21\ \Omega$

$Z_i = R_B \parallel Z_i' = 470\ k\Omega \parallel 816.21\ \Omega = \mathbf{814.8\ \Omega}$

(b) $A_v = \frac{-h_{fe}R_L}{h_{ie}+(h_{ie}h_{oe}-h_{fe}h_{re})R_L} = \frac{-(140)(22k\Omega)}{0.86k\Omega+((0.86k\Omega)(25\ \mu S)-(140)(15x10^{-4}))22k\Omega} = \mathbf{-357.68}$

(c) $A_i = \frac{I_O}{I_i} = \frac{h_{fe}}{1+h_{oe}R_L} = \frac{140}{1+(25\ \mu S)(22k\Omega)} = 132.70$

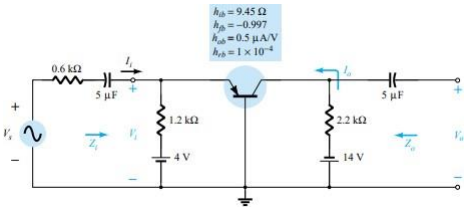
$$A_i' = \frac{I_o}{I_i'} = \left(\frac{I_o}{I_i}\right)\left(\frac{I_i}{I_i'}\right) \qquad I_i = \frac{470\ \text{k}\Omega\, I_i'}{470\ \text{k}\Omega + 0.816\ \text{k}\Omega}$$

$$= (132.70)(0.998) \qquad \frac{I_i}{I_i'} = 0.998$$

$$= \mathbf{132.43}$$

(d) $Z_o = \dfrac{1}{h_{oe} - (h_{fe} h re_f (h_{ie} + R_s)} = \dfrac{1}{(25\ \mu S) - ((140)(15x10^{-4}))(0.86k\Omega + 1k\Omega))} = \dfrac{1}{1371\mu S} = \mathbf{72.9k\Omega}$

26.



For the common-base amplifier of Fig. 8.84, determine:

(a) $Z_i' = h_{ib} - \dfrac{h_{fb} h_{rb} R_L}{1 + h_{ob} R_L} = 9.45k\Omega - \dfrac{(-0.997)(1x10^{-4})(2.2k\Omega)}{1 + (0.5\ \mu_V)(2.2k\Omega)} = 9.67\Omega$

$Z_i = 1.2\ k\Omega || Z' = 1.2\ k\Omega\ ||9.67\ \Omega = \mathbf{9.59\ \Omega}$

(b) $A_i' = \dfrac{h_{fb}}{1 + h_{ob} R_L} = \dfrac{-0.997}{1 + (0.5\ \mu_V)(2.2k\Omega)} = -0.996$

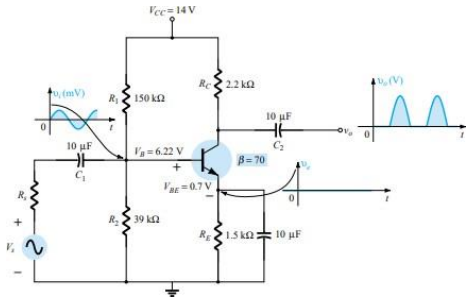
$A_i = \dfrac{I_o}{I_i} = \dfrac{I_o}{I_i'} \cdot \dfrac{I_i'}{I_i} = (-0.996)\dfrac{12\ k\Omega}{12\ k\Omega + 967\ k\Omega} = \mathbf{-0.998}$

(c) $A_v = \dfrac{-h_{fb} R_L}{h_{ib} + (h_{ib} h_{ob} - h_{fb} h_{rb}) R_L} = \dfrac{-(-0.997)(2.2k\Omega)}{9.45\Omega + ((9.45\Omega))(0.5\mu A/V) - (0.997)(1x10^{-4}))2.2k\Omega} = \mathbf{226.21}$

(d) $Z_o' = \dfrac{1}{h_{ob} - [h_{fb} h_{rb} / h_{ib}]} = \dfrac{1}{(0.5\mu A/V) - ((-0.997)(1x10^{-4}))/9.45\ \Omega} = 90.5k\Omega$

$Z_o = 2.2k || Z_o' = \mathbf{2.15k\Omega}$

27.



Given the network of Fig. 8.85:

- Determine if the system is operating properly based on the voltage-divider bias levels and expected waveforms for V_B and V_E.
- Determine the reason for the dc levels obtained and why the waveform for V_o was obtained.

28. Using PSpice Windows, determine the gain for the network of Fig. 8.6. Use Probe to display the input and output waveforms.
29. Using PSpice Windows, determine the gain for the network of Fig. 8.13. Use Probe to display the input and output waveforms.
30. Using PSpice Windows, determine the gain for the network of Fig. 8.25. Use Probe to display the input and output waveforms.